

Revised Programming Specification: Steam Review Analysis and Insights Tool

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General Project Description

The Steam Review Analysis and Insights Tool is designed to support User Research and Player Insights teams by collecting and analyzing Steam user reviews for video games. The project uses natural language processing, statistical analysis, and AI-assisted interpretation to identify patterns in player feedback that may be useful during game development.

The tool processes Steam review datasets by cleaning the data, performing sentiment analysis, categorizing reviews into gameplay-related themes, engineering additional features, generating visualizations, and running statistical analyses. AI-generated summaries are then used to explain the results of each analysis in plain language.

Rather than manually reading hundreds or thousands of reviews, researchers can quickly identify common player concerns, recurring themes, and statistically significant relationships within the dataset.

Revision to Original Scope

Original Plan

The original proposal included:

- A Flask/FastAPI web application
- User accounts and login
- Interactive search interface
- Dashboard for exploring results
- Database storage of scraped Steam data
- Scraping Steam Community discussions and reviews

Revised Plan

After implementing the core functionality and speaking with Prof. Harding, I determined that building both a web application and a comprehensive analytics system would not be feasible within the remaining course timeline.

The revised project focuses entirely on the backend analysis pipeline.

This allows more time to implement meaningful data analysis techniques while producing a cleaner and more maintainable project.

Current Project Workflow

1. Data Collection

Current implementation:

- Load Steam review datasets from CSV files.
- Remove duplicate reviews.
- Prepare the dataset for analysis.

Revision: Community discussion scraping has been postponed. The project currently analyzes exported Steam review datasets.

2. Natural Language Processing

Implemented:

- Sentiment analysis using TextBlob.
- Keyword-based categorization into gameplay-related themes.

Current categories include:

- Usability Issues
- New Mechanic Reception
- Competitive Analysis
- AI-Generated Content Complaints
- Monetization & Value
- Story / Campaign Sentiment

- Technical Requirements
 - SBMM / Matchmaking Frustration
 - Community & Social Sentiment
 - General Game Reception Sentiment
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3. Feature Engineering

Implemented:

- Review length
 - Word count
 - Sentiment strength
 - Boolean category indicators for statistical testing
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4. Statistical Analysis

Implemented:

- Descriptive statistics
- Histograms
- Category frequency bar charts
- Box plots
- Correlation matrices
- T-tests
- Chi-square tests
- Linear regression

These analyses help identify relationships between review characteristics, player sentiment, and discussion topics as they relate to one another.

5. AI-Assisted Interpretation

Implemented:

- Installed Ollama
- Added in automatic AI-generated summaries for:
 - Visualizations
 - Statistical analyses

- Regression output
- Correlation results

This provides human-readable explanations alongside the statistical outputs.

Current Program Flow

1. Load Steam review dataset.
 2. Clean and prepare the data.
 3. Perform sentiment analysis.
 4. Categorize reviews into gameplay themes.
 5. Engineer additional analytical features.
 6. Generate visualizations.
 7. Perform statistical analyses.
 8. Generate AI-assisted summaries of each result.
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Core Technologies

- Python
 - Pandas
 - TextBlob
 - NumPy
 - Matplotlib
 - SciPy
 - Statsmodels
 - Ollama (Llama 3.2)
 - Jupyter Notebook / VS Code
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Features Completed

- CSV review loading
- Data cleaning
- Sentiment analysis
- Keyword categorization

- Feature engineering
 - Descriptive statistics
 - Histograms
 - Bar charts
 - Box plots
 - Correlation analysis
 - Correlation heat maps
 - T-tests
 - Chi-square tests
 - Linear regression
 - AI-generated statistical summaries
 - Automated analysis pipeline
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Features Removed or Deferred

The following features have been removed from the required scope to make the project feasible within the remaining course timeline:

- Flask/FastAPI web application
- User login system
- Interactive dashboard
- Database storage
- Steam Community discussion scraping
- Report exporting through a web interface

These may be revisited in future iterations of the project.

Proposed "Nice to Have" Features (Time Permitting)

If time allows after completing the core project, I would like to implement:

- Compare multiple games in a single analysis.
- Export automated PDF reports for stakeholder presentations.
- Improve keyword categorization using machine learning.
- Add additional statistical measures such as effect sizes and confidence intervals.
- Improve visualizations with interactive charts.

- Compare sentiment trends across multiple review datasets.
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Updated Technical Flow

Data Layer

- Load Steam review CSV files
- Remove duplicate reviews

Natural Language Processing

- Sentiment analysis
- Keyword categorization

Feature Engineering

- Review length
- Word count
- Sentiment strength
- Category indicators

Statistical Analysis

- Descriptive statistics
- Correlation analysis
- T-tests
- Chi-square tests
- Linear regression

Visualization

- Histograms
- Bar charts
- Box plots
- Correlation heat maps

AI Interpretation

- Automatic Ollama summaries for every visualization and statistical analysis
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Revised Self Assessment

The biggest change from my original program spec was recognizing that the analytical component of the project was significantly more valuable than building a front-end application. Rather than trying to complete two substantial projects simultaneously, it made more sense to narrow the scope to focus on developing a reusable backend analysis pipeline, per conversations with Prof. Harding.

This decision allowed me to implement more advanced statistical analyses, improve the organization of the codebase, and integrate AI-assisted interpretation. I believe the revised project is both more technically interesting and more achievable within the remaining weeks of the course.

I am confident that I can complete the revised scope. The core functionality is now operational, and the remaining work focuses primarily on expanding the analyses, improving documentation, refining AI summaries, and polishing the overall presentation.